

Qatar-4b

R-2591

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-4_201106 · created 2026-08-27T23:20:16+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459159.6942 +/- 0.0024 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.087 +/- 0.05
Transit depth (Rp/Rs)^2	0.75 +/- 0.87 [%]
Orbital Inclination (inc)	82.57 +/- 2.19
Airmass coefficient 1 (a1)	25.22 +/- 18.99
Airmass coefficient 2 (a2)	-2.98 +/- 1.05
Scatter in the residuals of the lightcurve fit is	85.49 %
Best Comparison Star	None
Optimal Aperture	3.89
Optimal Annulus	11.69
Transit Duration (day)	0.041 +/- 0.029

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.087 ± 0.05 vs 0.138 (deviation 0.28σ)
Duration fit vs published	0.041 d vs 0.0892 d (deviation 0.46σ)
Mid-time O–C	-4.0 min
Depth SNR	0.5
Residual scatter	85.49 %

Light curve

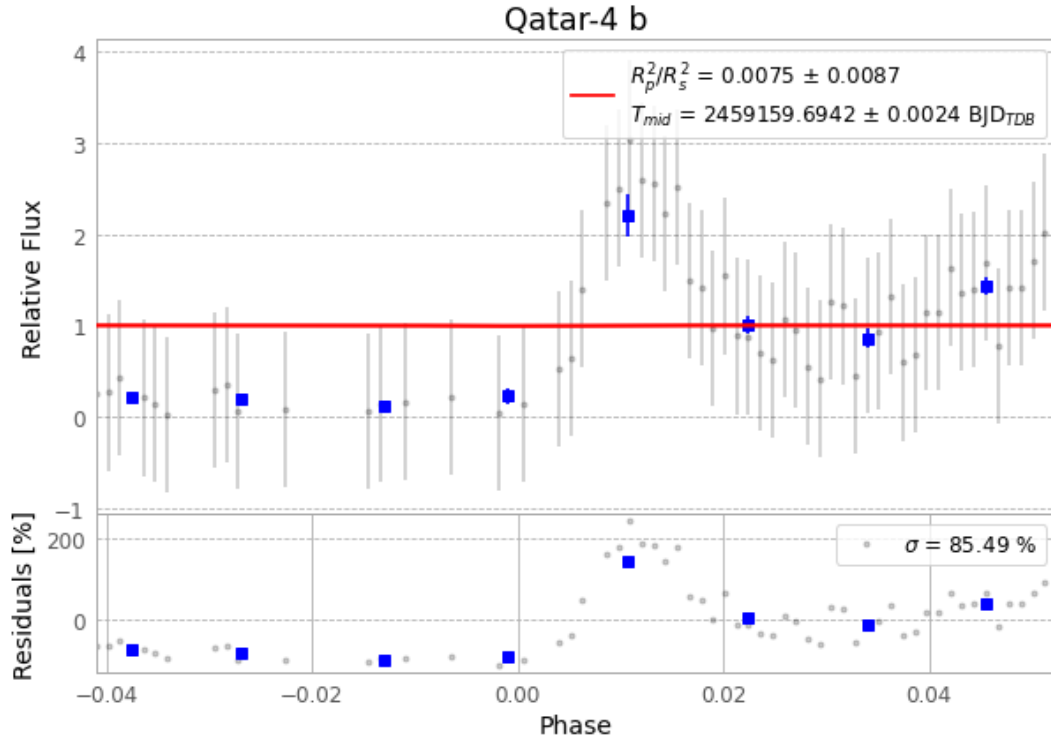


Figure 1 — FinalLightCurve_Qatar-4 b_2020-11-05.png (SHA-256 8bc69da9524e162a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [522, 314], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 2e96d1a86ef65098...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

82 FITS frames (54 MB), Qatar-4201106024512.FITS → Qatar-4201106064812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-4-201106.log (67a7d26e204f...), FinalLightCurve_Qatar-4 b_2020-11-05.png (8bc69da9524e...), FinalLightCurve_Qatar-4 b_2020-11-05.csv (9c9fb54517a4...)

Compute resources

Wall time 148.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-27 23:20Z.